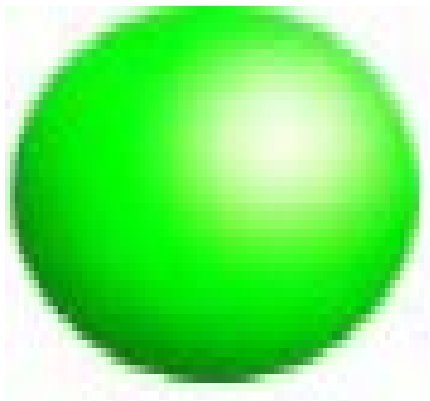


PROJECTION OF SOLIDS

INTRODUCTION

A solid has three dimensions, viz. length, breadth and thickness. To represent a solid on a flat surface having only length and breadth, at least two orthographic views are necessary.

SOLIDS OF REVOLUTION



SPHERE

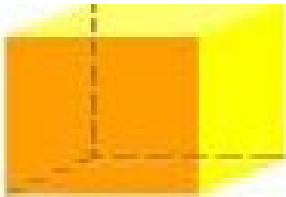


CYLINDER

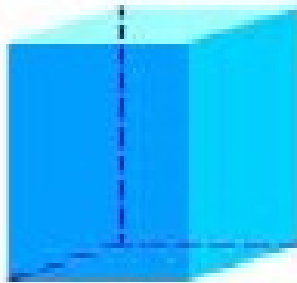


CONE

POLYHEDRONS



CUBE



PRISM



PYRAMID

Overview

1. Types of solids.
2. Projections of solids in simple positions.
 - a. Axis perpendicular to the H.P.
 - b. Axis perpendicular to the V.P.
 - c. Axis parallel to both the H.P. and the V.P
3. Projections of solids with axes inclined to one of the reference planes and parallel to the other.
 - a. Axis inclined to the V.P. and parallel to the H.P.
 - b. Axis inclined to the H.P. and parallel to the V.P.
4. Projections of solids with axes inclined to both the H.P. and the V.P.
5. Projections of spheres.

Types of Solids

Solids may be divided into two main groups:

i. Polyhedron

ii. Solids of revolution.

i. Polyhedron: A polyhedron is defined as a solid bounded by planes called faces. When all the faces are equal and regular, the polyhedron is said to be regular.

There are six regular polyhedron

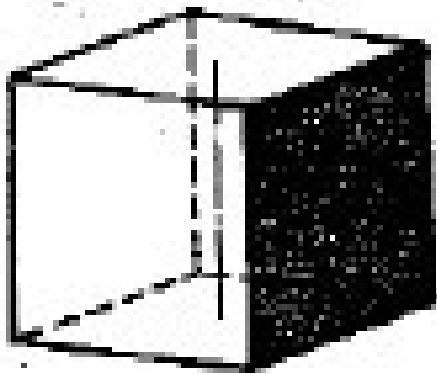
- **Tetrahedron :**
It has four equal faces, each an equilateral triangle.



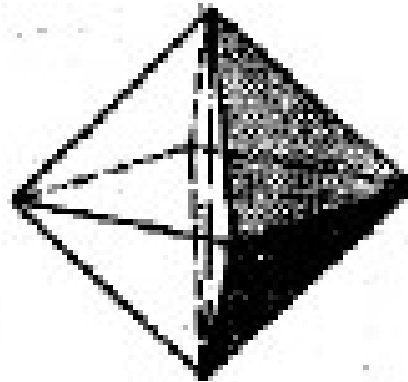
b. **Cube or Hexahedron** : It has six faces, all equal squares.

c. **Octahedron** : It has eight equal equilateral triangles as faces.

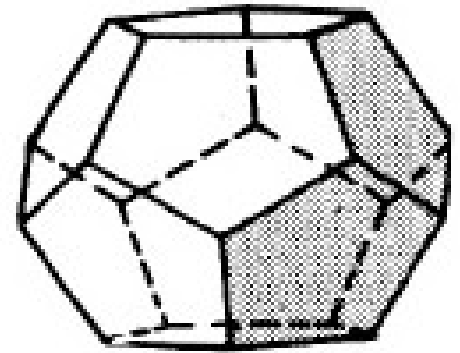
d. **Dodecahedron** : It has twelve equal and regular pentagons as faces.



Cube



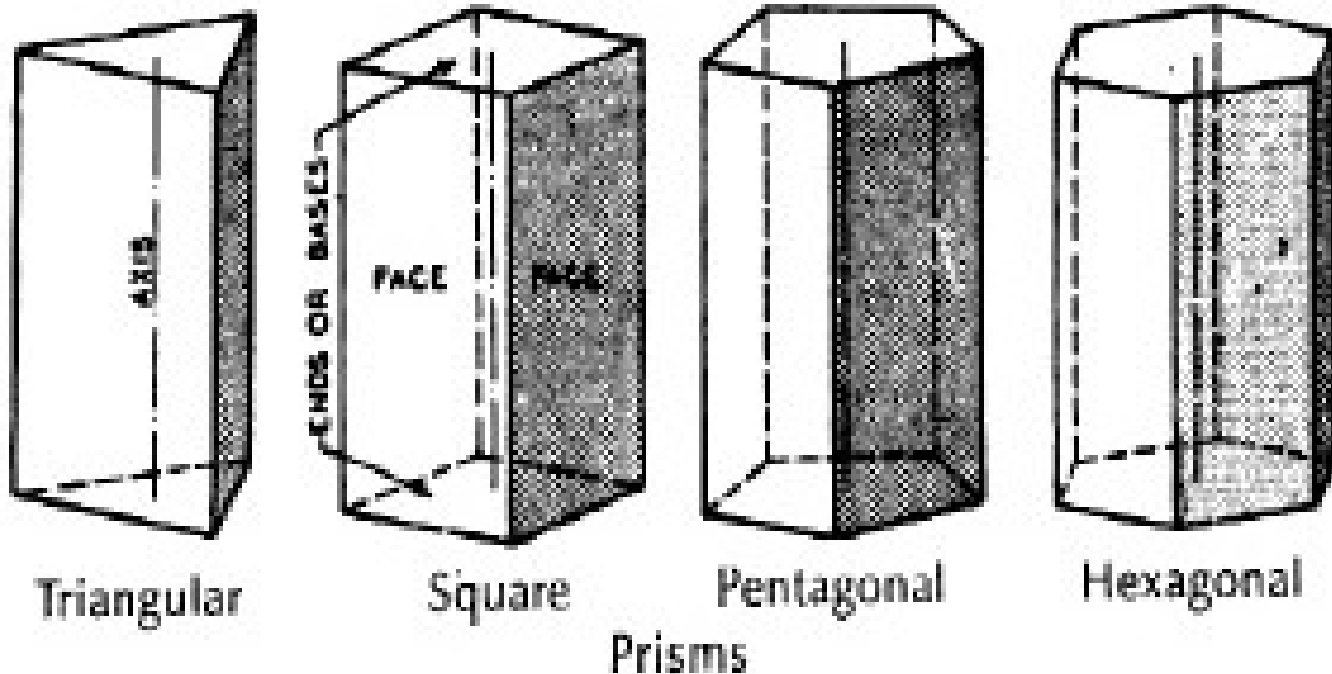
Octahedron ✓



Dodecahedron

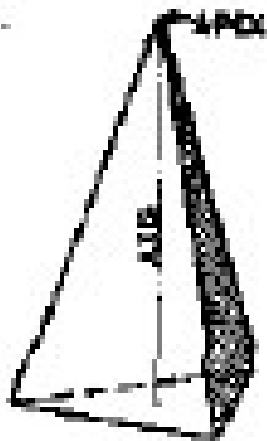
Prism: This is a polyhedron having two equal and similar faces called its ends or bases, parallel to each other and joined by other faces, which are parallelograms. The imaginary line joining the centers of the bases is called the axis.

A right and regular prism has its axis perpendicular to the bases. All its faces are equal rectangles.

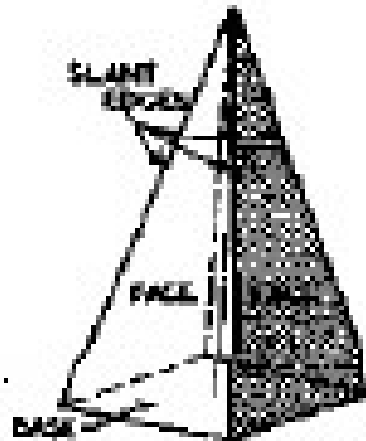


Pyramid: This is a polyhedron having a plane figure as a base and a number of triangular faces meeting at a point called the vertex or apex. The imaginary line joining the apex with the centre of the base is its axis.

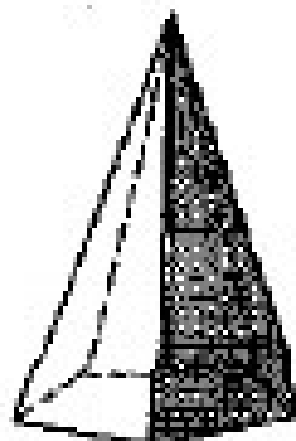
A right and regular pyramid has its axis perpendicular to the base, which is a regular plane figure. Its faces are all equal isosceles triangles.



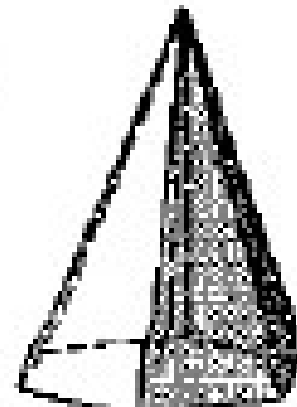
Triangular



Square



Pentagonal



Hexagonal

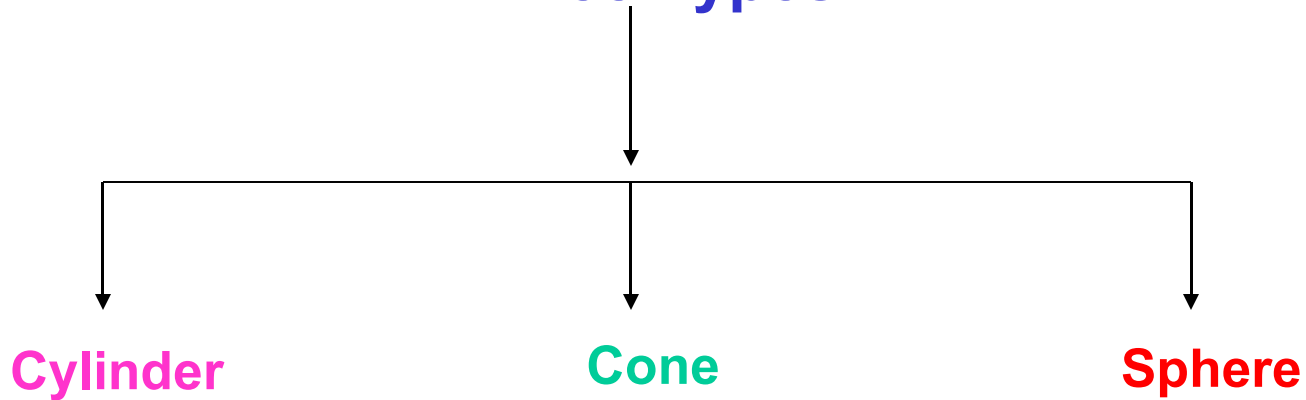
Pyramids

Solids of Revolution

Definition:

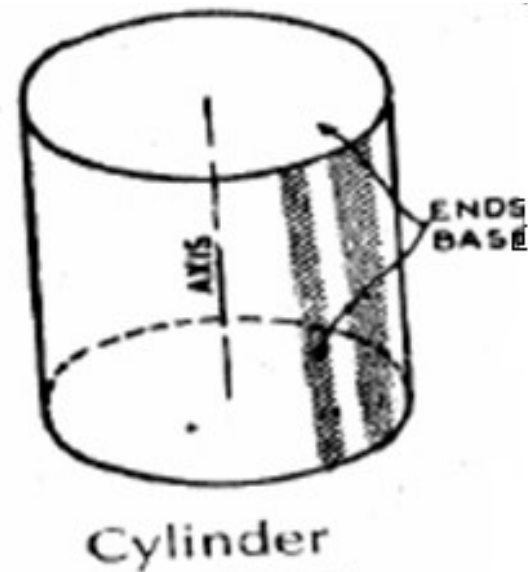
Solids which are generated by rotating a plane surface about one of its sides

Three Types



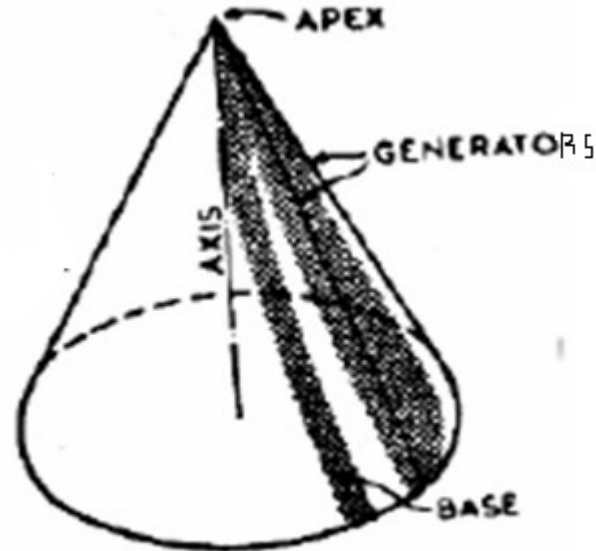
a) Cylinder

A right circular cylinder is a solid generated by the revolution of a rectangle about one of its sides, which remains fixed. It has two equal circular bases. The line joining the centers of the bases is the axis. It is perpendicular to the bases.



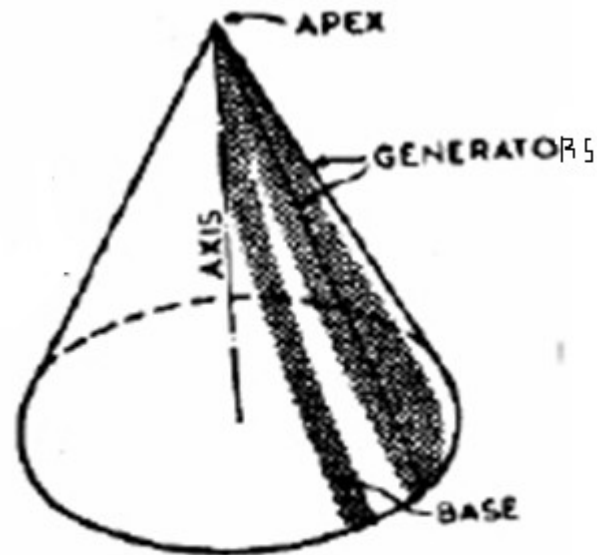
b. Cone :

A right circular cone is a solid generated by the revolution of right-angled triangles about one of its perpendicular sides, which is fixed.



Cone

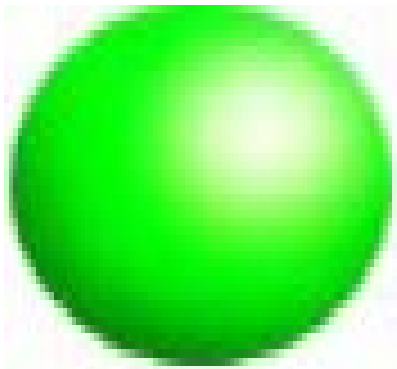
It has one circular base. Its axis joins the apex with the centre of the base to which it is perpendicular. Straight lines drawn from the apex to the circumference of the base-circle are all equal and are called generators of the cone.



Cone

c. Sphere :

A sphere is a solid generated by the revolution of a semi-circle about its diameter as the axis. The mid-point of the diameter is the centre of the sphere. All points on the surface of the sphere are equidistant from its centre.

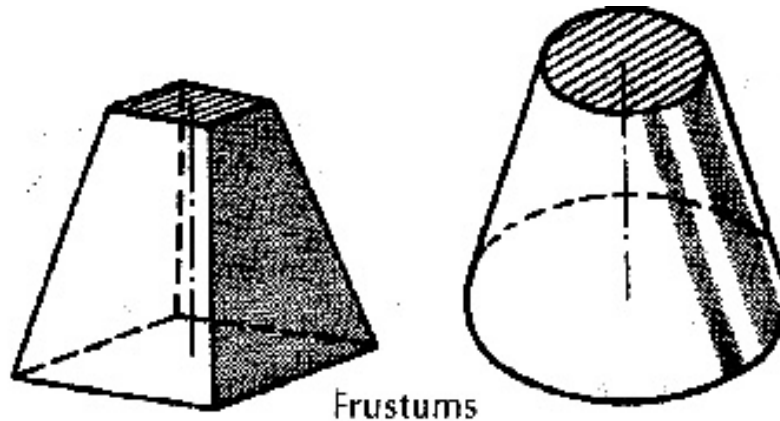


Sphere

d. **Frustum:** When a pyramid or a cone is cut by a plane parallel to its base, thus removing the top portion, the remaining portion is called its frustum.

e. **Truncated:** When a solid is cut by a plane inclined to base it is said to be truncated.

Unless and otherwise stated, the given solid should be understood as right and regular.



Projections of Solids in Simple Positions

- A solid in simple position may have its axis perpendicular to one reference plane or parallel to both.
- When the axis is perpendicular to one reference plane, it is parallel to the other. Also, when the axis of a solid is perpendicular to a plane, its base will be parallel to that plane.
- We have already seen that when a plane is parallel to a reference plane, its projection on that plane shows its true shape and size.

Therefore, the projection of a solid on the plane to which its axis is perpendicular will show the true shape and size of its base.

Hence, when the axis is perpendicular to the ground, i.e. to the H.P., the top view should be drawn first and the front view projected from it.

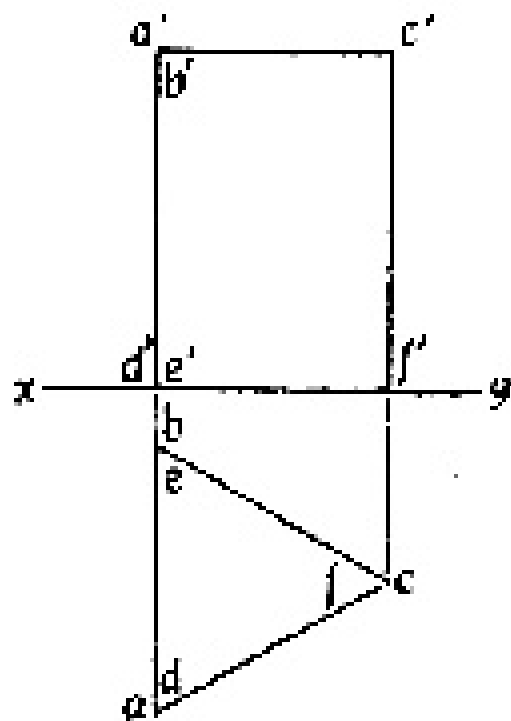
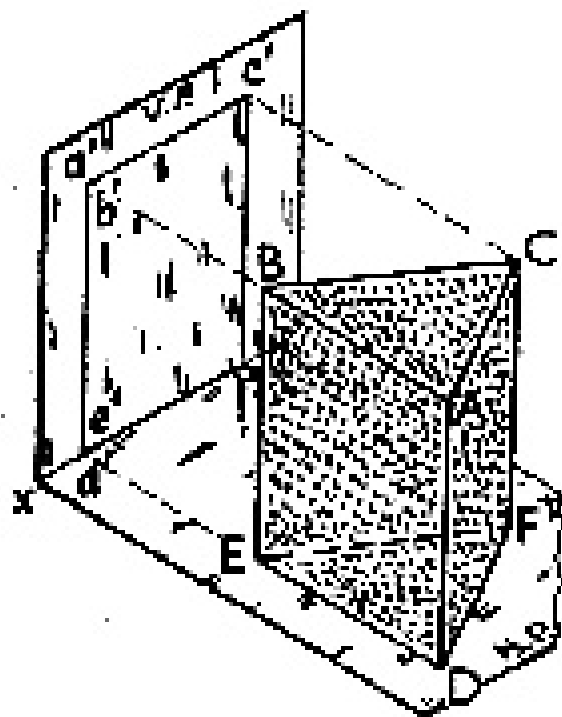
When the axis is perpendicular to the V.P., beginning should be made with the front view. The top view should then be projected from it.

When the axis is parallel to both the H.P. and the V.P., neither the top view nor the front view will show the actual shape of the base. In this case, the projection of the solid on an auxiliary plane perpendicular to both the planes, viz. the side view must be both the planes, viz. the side view must be drawn first. The front view and the top view are then projected from the side view. The projections in such cases may also be drawn in two stages.

Axis perpendicular to the H.P.

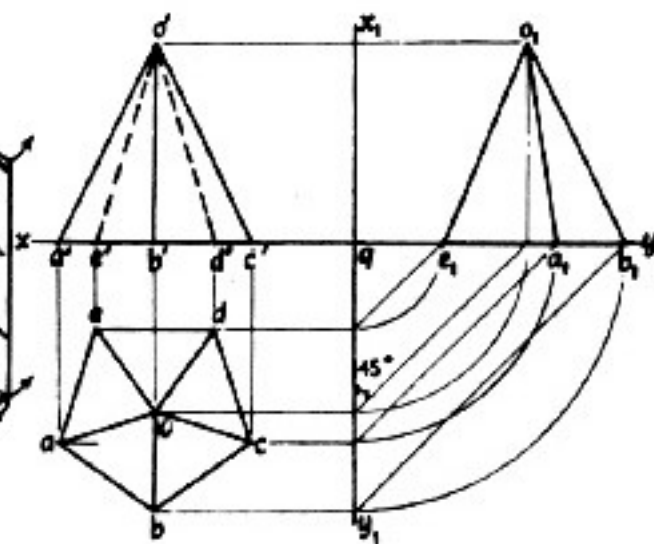
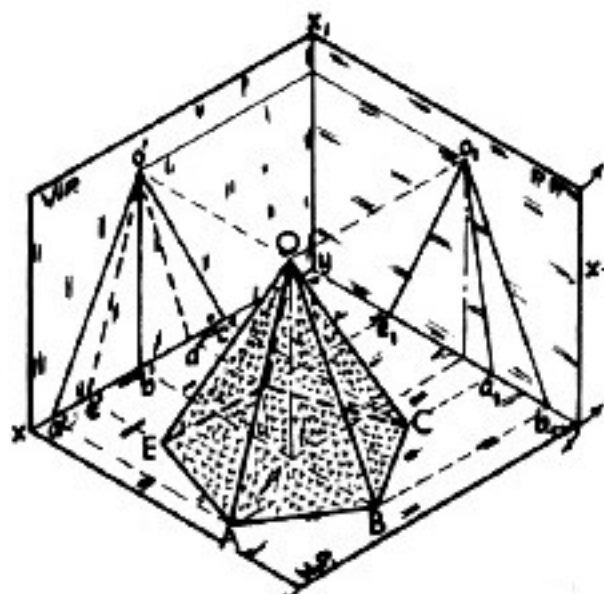
Problem 1 : Draw the projections of a triangular prism, base 40 mm side and axis 50 mm long, resting on one of its bases on the H.P. with a vertical face perpendicular to the V.P.

- ii. As the axis is perpendicular to the ground i.e. the H.P. begin with the top view. It will be an equilateral triangle of sides 40 mm long with one of its sides perpendicular to xy . Name of corners as shown, thus completing the top view. The corners d , e and f are hidden and coincide with the top corners a , b and c respectively.
- ii. Project the front view, which will be a rectangle. Name the corners. The line $b'e'$ coincides with $a'd'$.



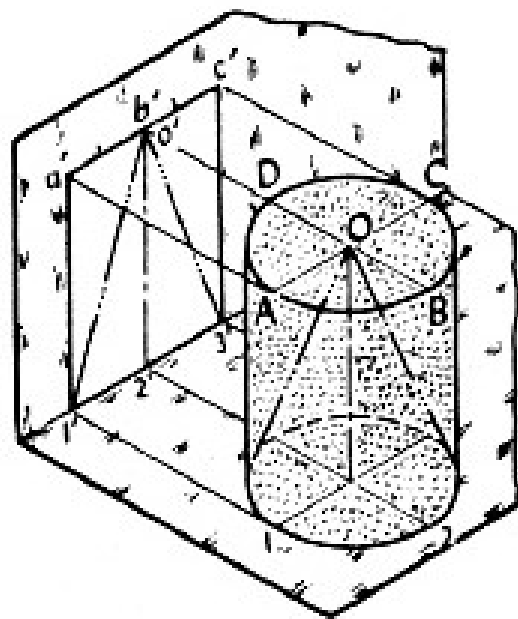
Problem 2. :

Draw the projections of a pentagonal pyramid, base 30 mm edge and axis 50 mm long, having its base on the H.P. and an edge of the base parallel to the V.P. Also draw its side view.

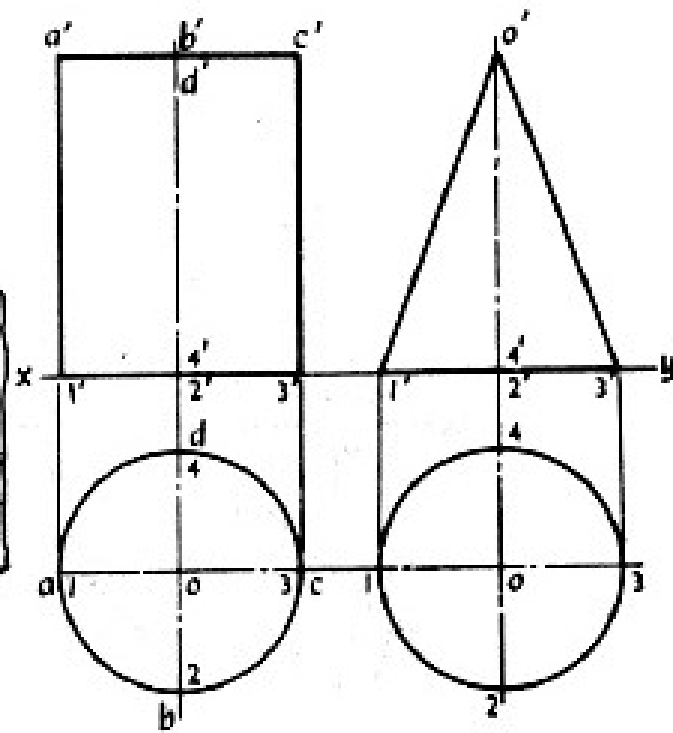


Problem 3.

Draw the projections of (i) a cylinder, base 40 mm diameter and axis 50 mm long, and (ii) a cone, base 40 mm diameter and axis 50 mm long, resting on the H.P. on their respective bases.



(i)

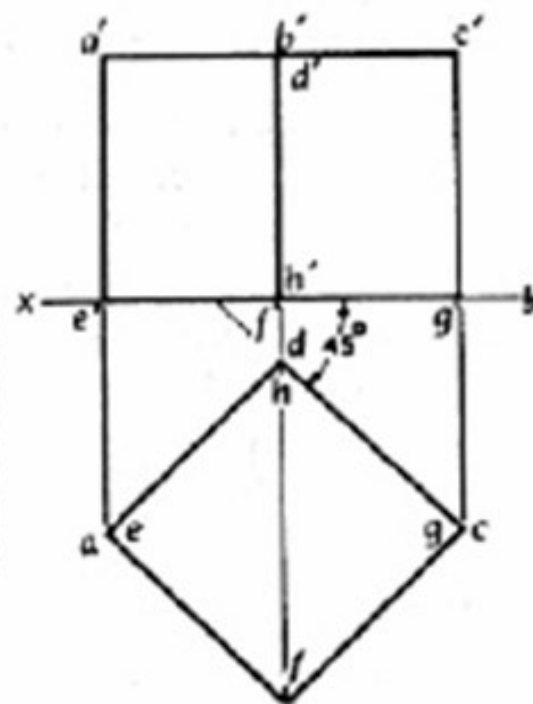
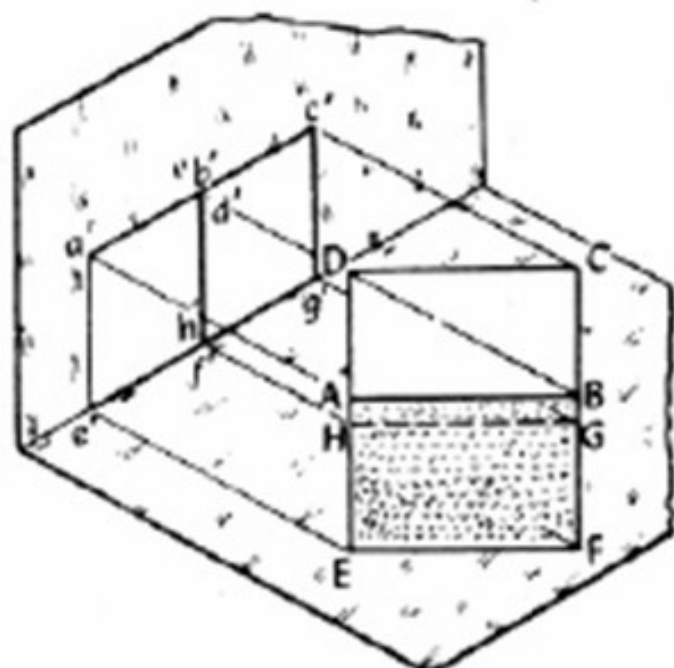


(ii)

(iii)

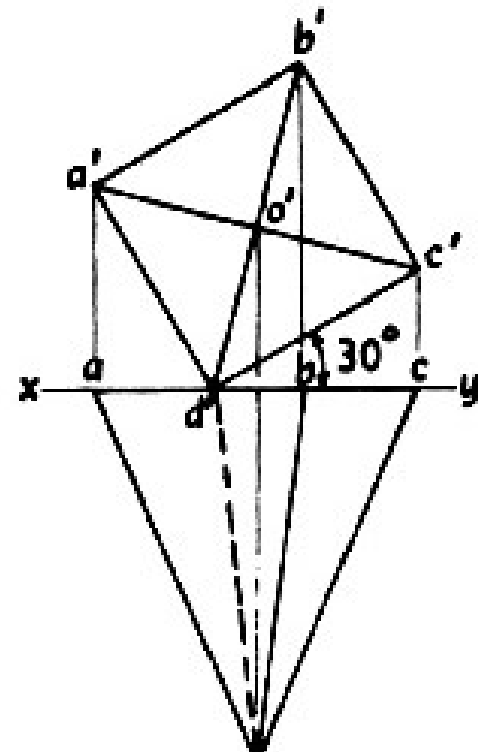
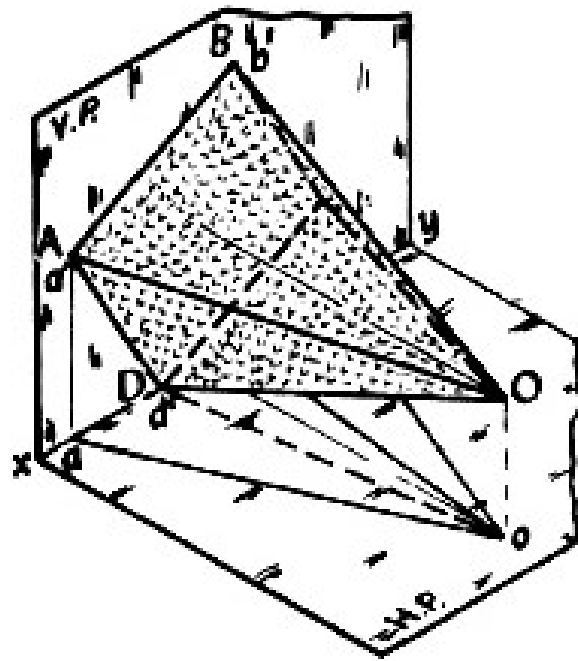
Problem 4.

A cube of 50 mm long edges is resting on the H.P. with its vertical faces equally inclined to the V.P. Draw its projections.



Problem 5.

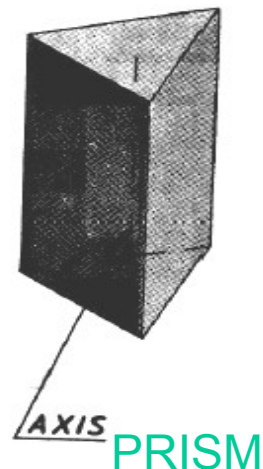
A square pyramid, base 40 mm side and axis 65 mm long, has its base in the V.P. One edge of the base is inclined at 30 degrees to the H.P. and a corner contained by that edge is on the H.P. Draw its projections.



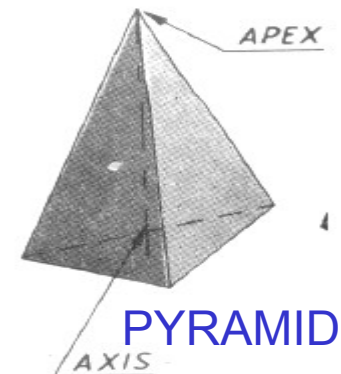
Projections of Solids with axes inclined to one of the reference planes and parallel to the other plane

When a solid is placed on the HP with axis inclined to horizontal plane of projection, the elemental portion of the solid that lies on H.P depends upon the type of solid.

When a prism is placed on H.P with its axis inclined to it, then it will lie either on one of its base edges or one of its corners on H.P

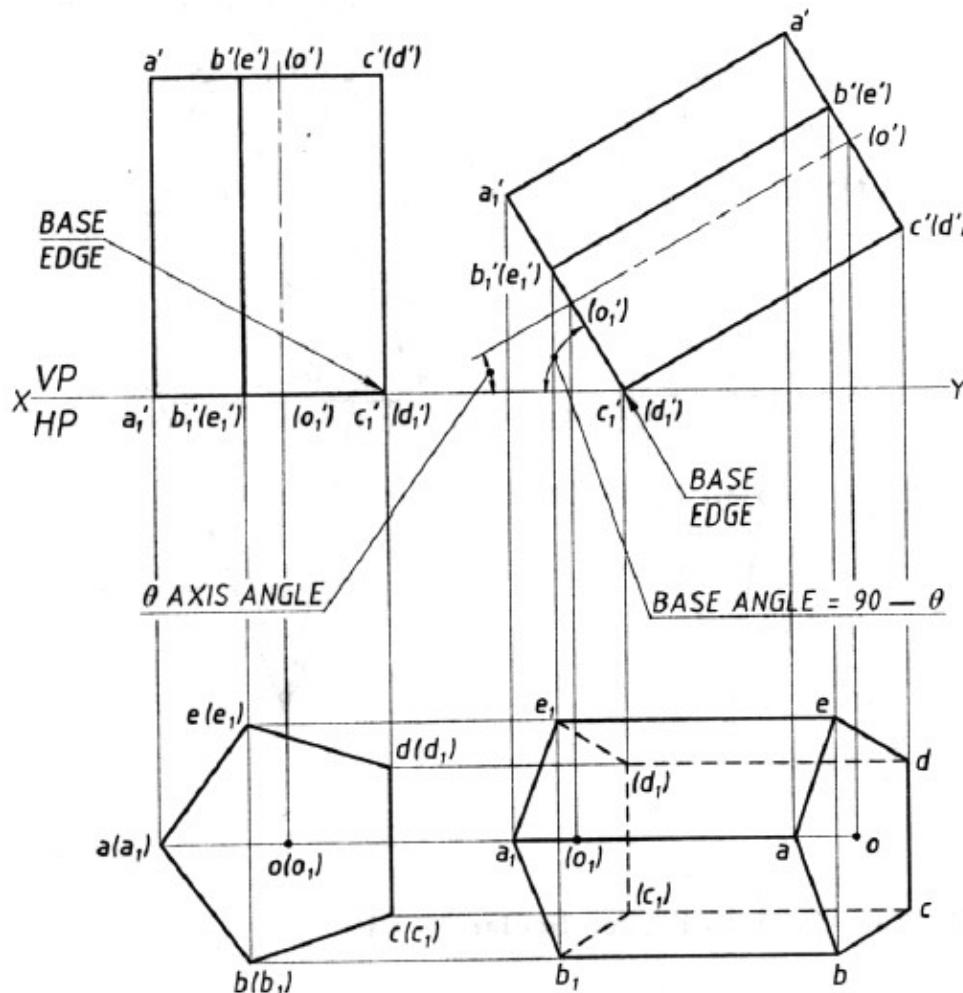


When a pyramid is placed on H.P with its axis inclined to H.P then we will have one of its base edges on H.P (or) one of its base corners on H.P (or) one of its slant edges on H.P (or) one of its triangular faces on H.P (or) an apex on H.P.



When a solid lies with an edge of the base on the H.P

If the solid is required to be placed with an edge of the base on the H.P then initially the solid has to be placed with its base on H.P such that an edge of the base is perpendicular to V.P i.e to XY line in the top view preferably on the right side.



When a pentagonal prism has to be placed with an edge of base on H.P such that the base or axis is inclined to H.P then initially the prism is placed with its base on H.P with an edge of its base perpendicular to V.P and lying to the right side.

In this position the first set of top views and front views are drawn with the base edges $(c_1)(d_1)$ perpendicular to XY line in the top view.

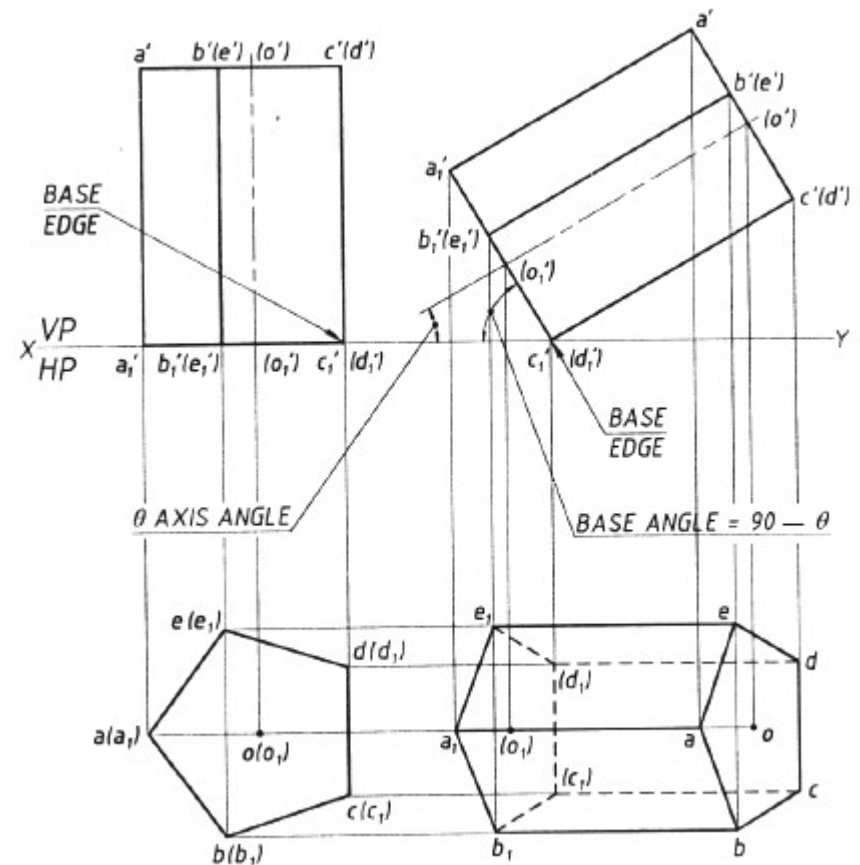
In the front view this edge $c_1'(d_1')$ appears as a point.

Since the prism has to lie with an edge of the base on the H.P the front view of the prism is tilted on the edge $c_1'(d_1')$ such that the axis is inclined at θ to H.P.

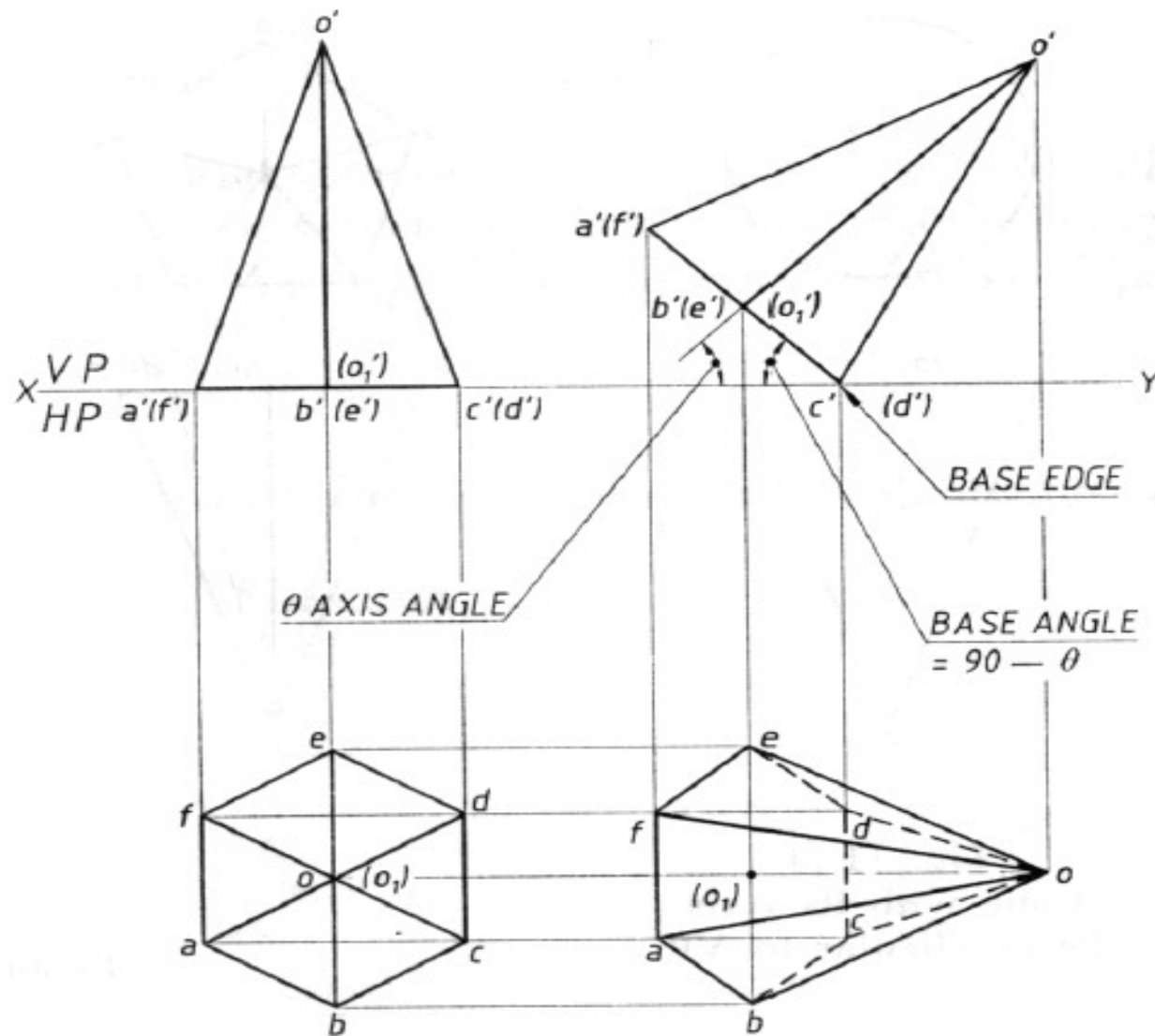
Redraw the first front view in the tilted position.

Whenever the inclination of axis θ with H.P is given first the base is drawn at $(90 - \theta)$ in the front view, otherwise improper selection of the position of the axis may result in the base edge $c_1'(d_1')$ lying above or below the XY line.

The second top view is projected by drawing the vertical projectors from the corners of the second front view and the horizontal projectors from the first top view.



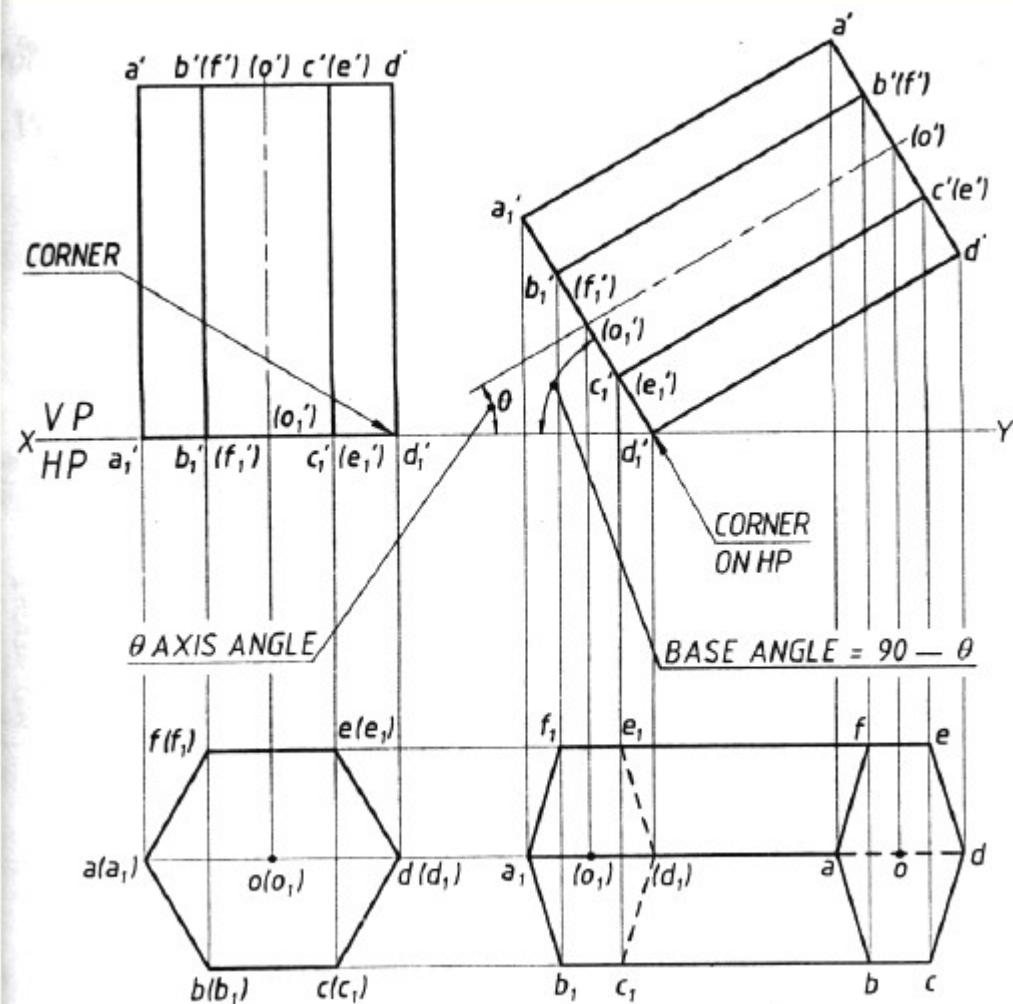
Method of obtaining the **top** and the **front** views of the **pyramid** when it lies on HP on one of its base edges with its axis or the base inclined to HP.



When the solid lies on one of its corners of the base on HP

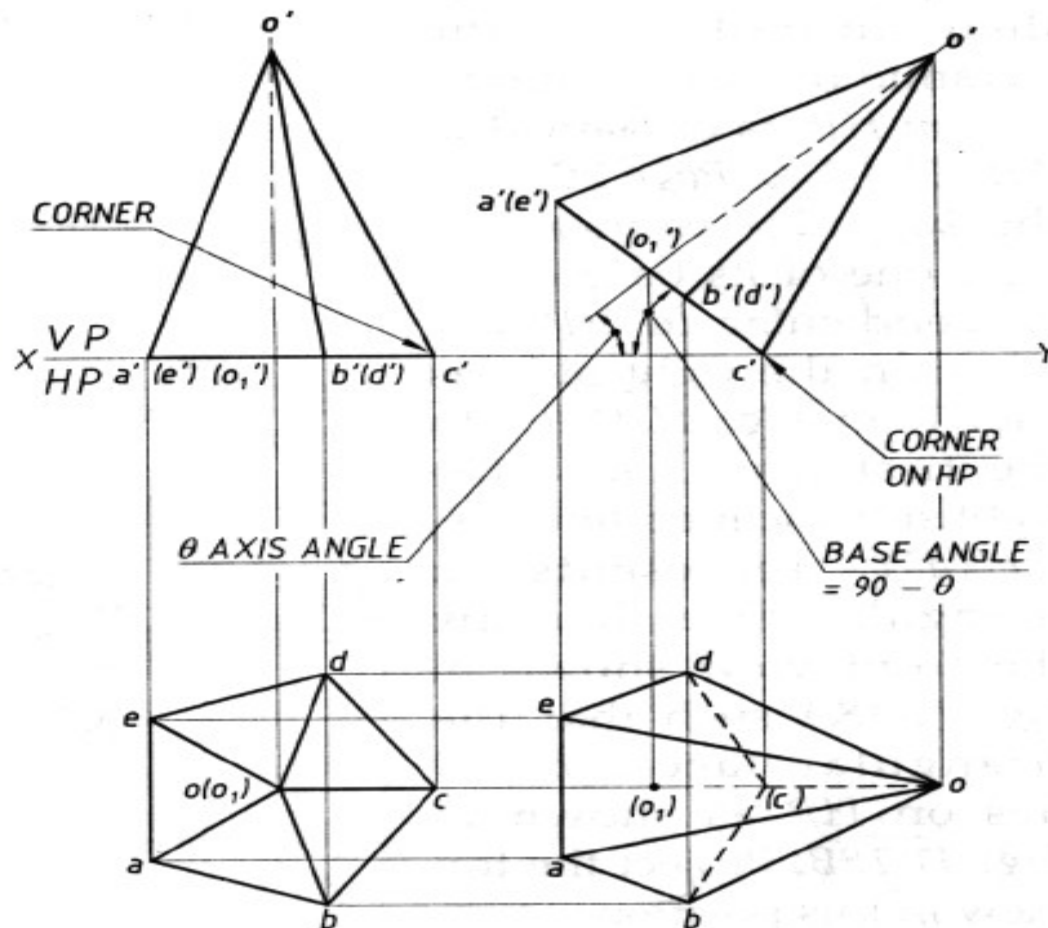
- When a solid lies on one of its corners of the base on HP, then the two edges of the base containing the corner on which it lies make either **equal inclinations** or **different inclination** with HP.
- **Corner of the base on HP with two base edges containing the corner on which it rests make equal inclinations with HP.**
- Initially the solid should be placed with its base on HP such that **an imaginary line connecting the center of the base and one of its corners is parallel to VP**, i.e. to XY line in the top view, and preferably to lie on the right side.

For example, when a **hexagonal prism** has to be placed with a **corner of the base on HP** such that the **base or the axis is inclined to HP**, then initially the prism is placed with its base on HP such that an **imaginary line connecting the center of the base and a corner is parallel to VP** and it lies on the **right side**.



➤ In this position, the first set of top and front views are drawn – the line $(o_1)(d_1)$ is parallel to the **XY** line in the top view.

Methods for obtaining top and front views of the pyramid when it rests on HP on one of its base corners such that the two base edges containing the corner on which it rests make equal inclinations with HP.

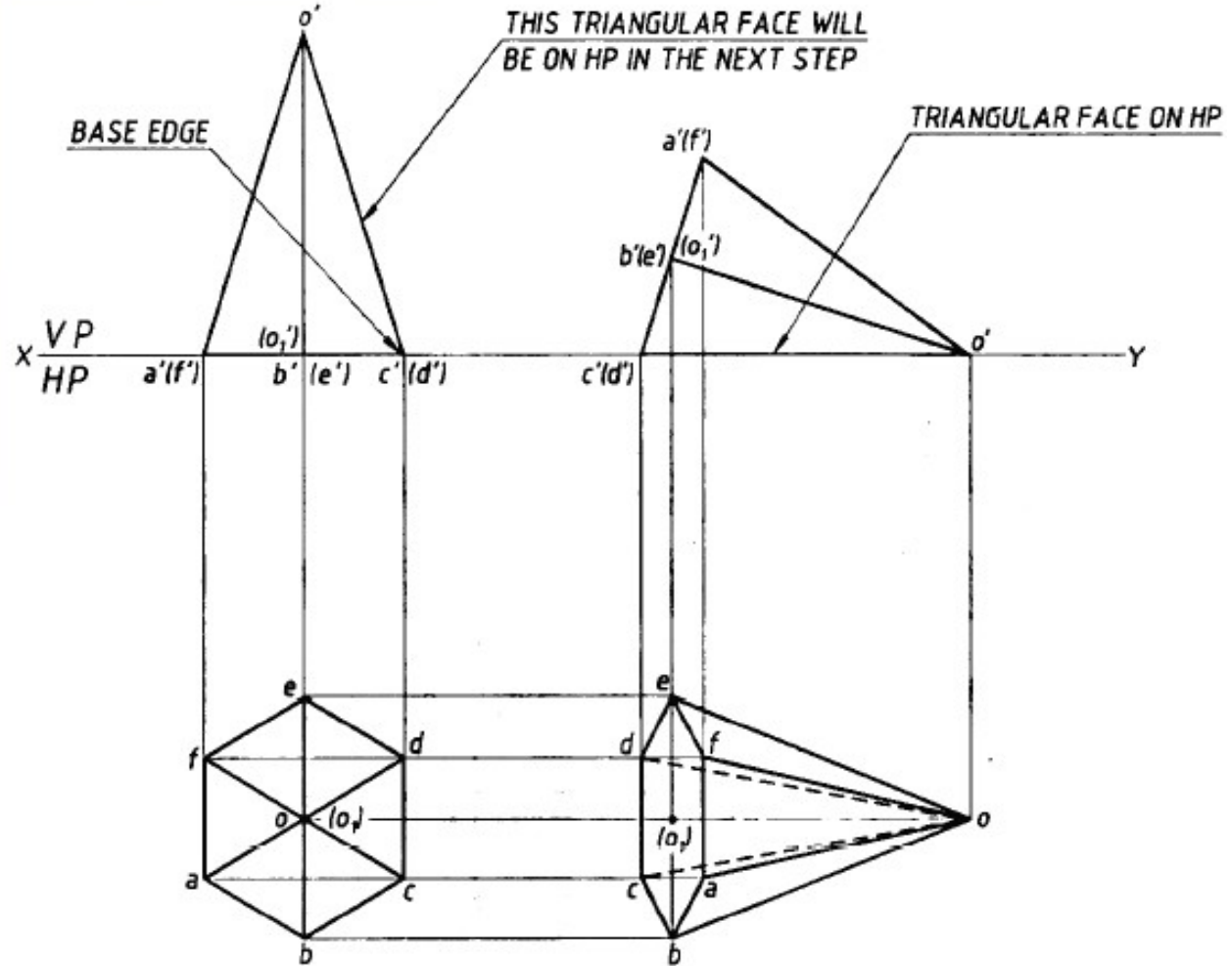


When a pyramid lies on one of its triangular faces on HP

If a pyramid has to be placed on one of its triangular faces on HP, then initially let the pyramid be placed with its base on HP.

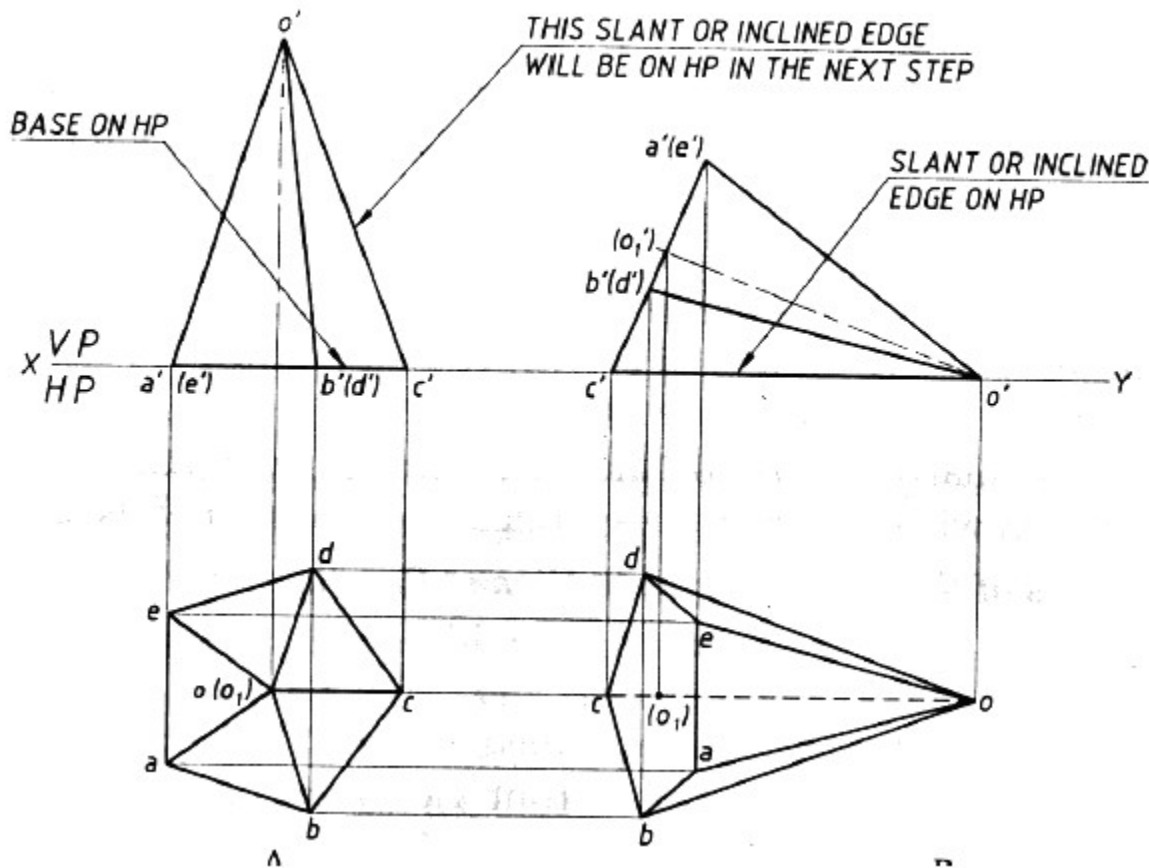
In the first front view, the right side inclined line, i.e., $o'c'(d')$ represents a triangular face.

Redraw the front view such that the triangular face $o'c'(d')$ lies on HP.



When a pyramid lies on one of its slant edges on HP

When a pyramid lies with one of its slant edges on HP, then two triangular faces containing the slant edge on which it rests make either **equal inclinations** or **different inclinations** with HP.



Projections of solids with axes inclined to one of the reference planes and parallel to the other

When a solid has its axis inclined to one plane and parallel to the other, its projections are drawn in two stages.

a. In the initial stage, the solid is assumed to be in simple position, i.e. its axis perpendicular to one of the planes.

If the axis is to be inclined to the ground, i.e. the H.P., it is assumed to be perpendicular to the H.P. in the initial stage.

Similarly, if the axis is to be inclined to the V.P., it is kept perpendicular to the V.P. in the initial stage.

Moreover

i. If the solid has an edge of its base parallel to the H.P. or in the H.P. or on the ground, that edge should be kept perpendicular to the V.P.; if the edge of the base is parallel to the V.P. or in the V.P., it should be kept perpendicular to the H.P.

ii. If the solid has a corner of its base in the H.P. or on the ground, the sides of the base containing that corner should be kept equally inclined to the V.P.; if the corner is in the V.P., they should be kept equally inclined to the H.P.

Methods of drawing the projections of solids

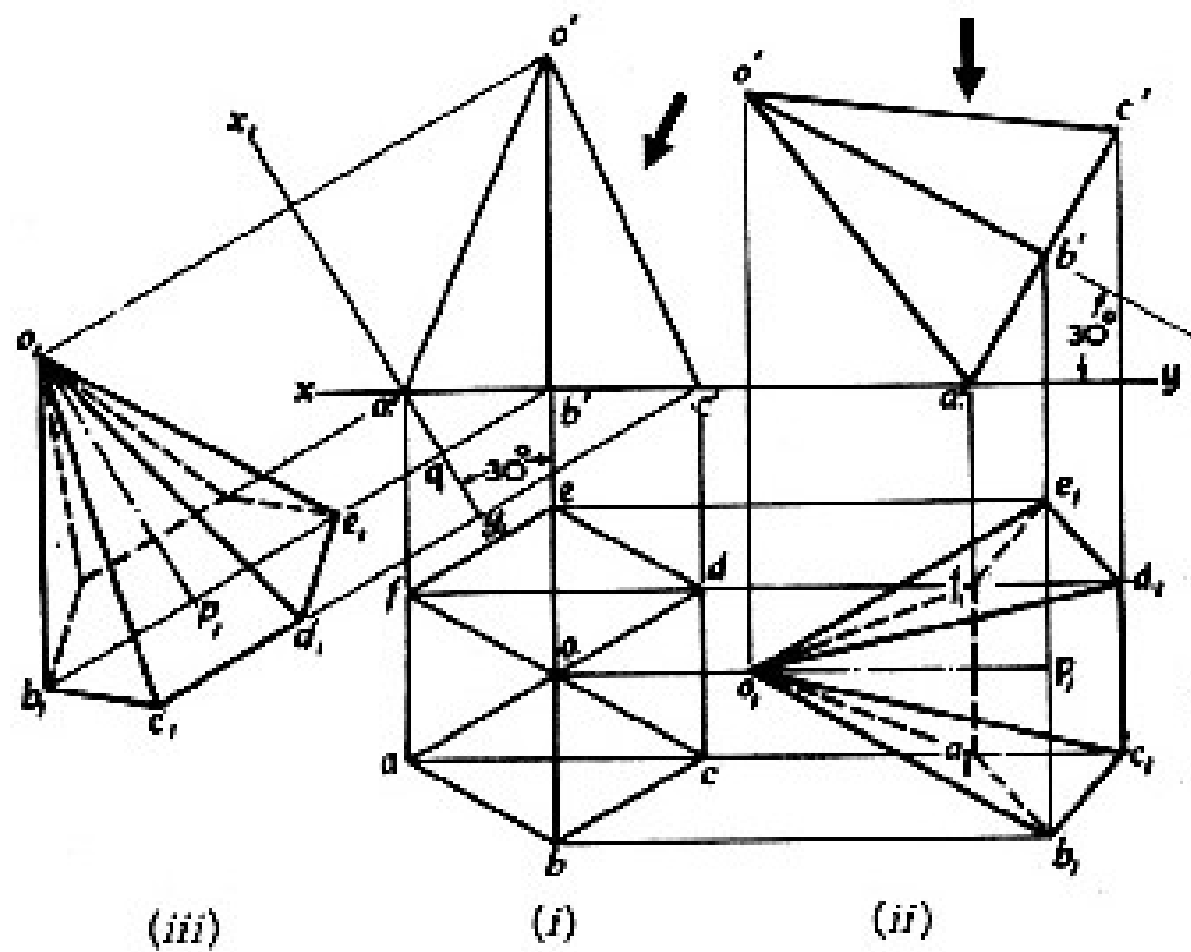
Two methods

Change of position method –

The solids are placed first in the simple position and then tilted successively in two or three stages to obtain the final position.

Auxiliary plane method (Change of reference-line method) –

The solids are placed initially in the simple position and then one or two auxiliary planes are setup to obtain the views in the required position.



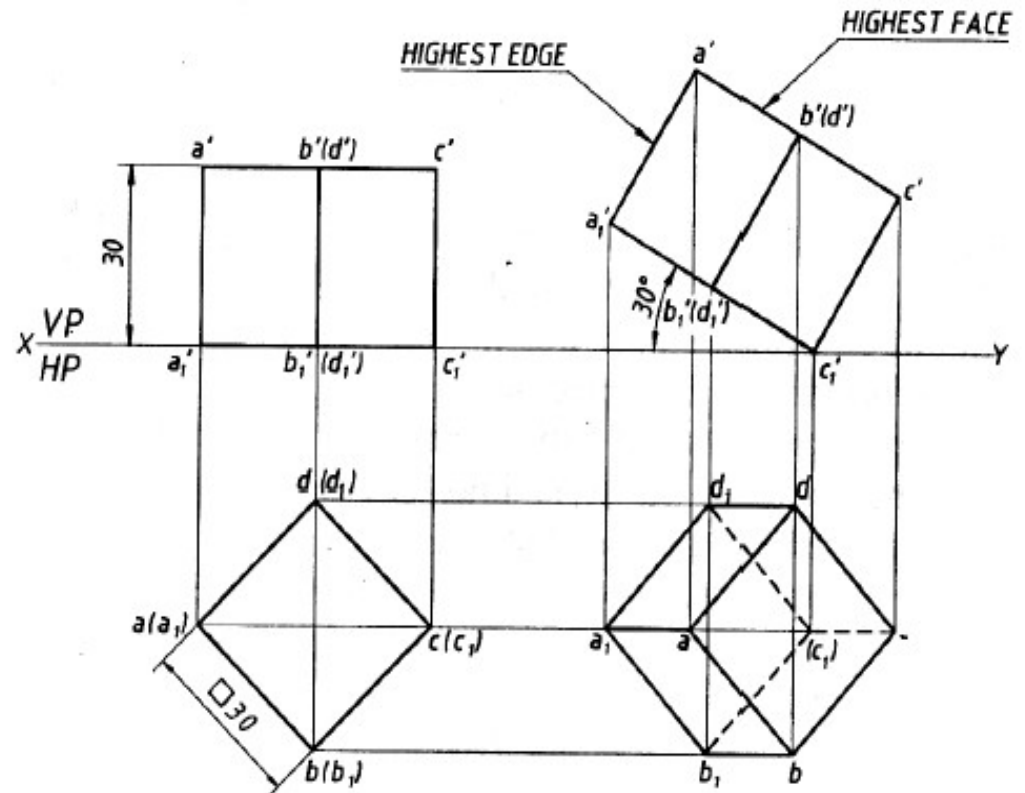
Problem

A cube of **30 mm** sides is held on one of its corners on **HP** such that the bottom square face containing that corner is inclined at **30°** to **HP**. Two of its adjacent base edges containing the corner on which it rests are equally inclined to **VP**. Draw the top and front views of the cube.

Solution:

The cube rests with one of its corners on **HP** such that the bottom square face containing the corner on which it rests is inclined at **30°** to **HP**.

First, the cube is assumed to lie with one of its faces completely on **HP** such that the two vertical faces make equal inclinations with **VP**.



➤ Draw a square **abcd** to represent the top view of the cube such that two of its sides make equal inclinations with the XY line, i.e., with VP.

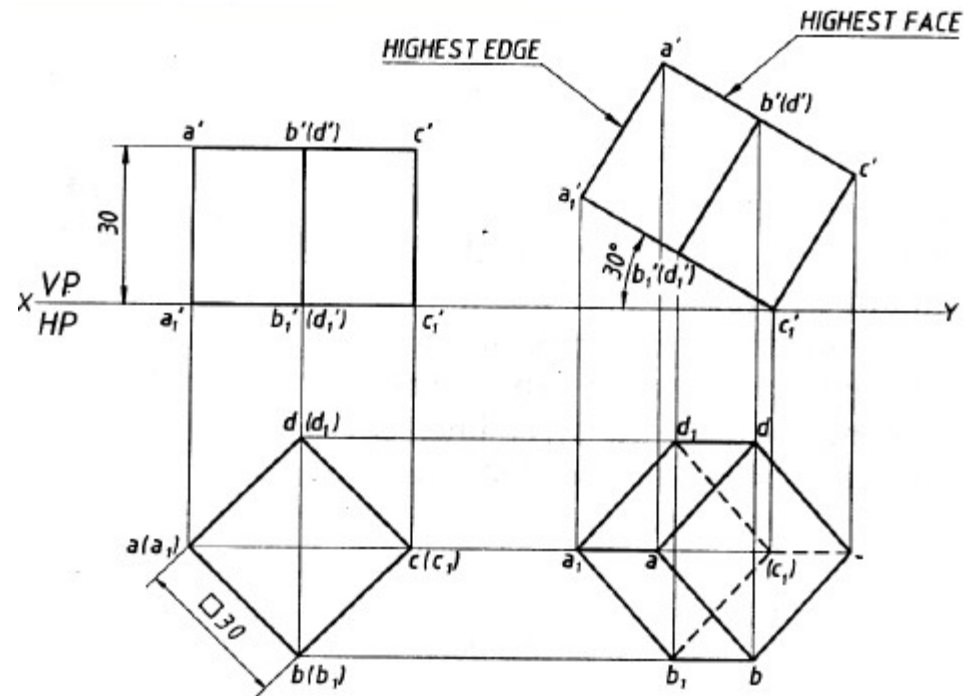
➤ Let (**a₁**), (**b₁**), (**c₁**) and (**d₁**) be the four corners of the bottom face of the cube which coincide in the top view with the corners **a**, **b**, **c** and **d** of the top face.

➤ Project the front view of the cube. The bottom face **a₁'b₁'c₁'(d₁)** in the front view coincide with the **XY** line.

➤ Now the cube is tilted on the bottom right corner **c₁'** such that the bottom face **a₁'b₁'c₁'(d₁)** is inclined at 30° to HP. Reproduce the front view with face **a₁'b₁'c₁'(d₁)** inclined at 30° to the XY line.

Draw the vertical projectors through all the corners in the reproduced front view and horizontal projectors through the corners of the first top view.

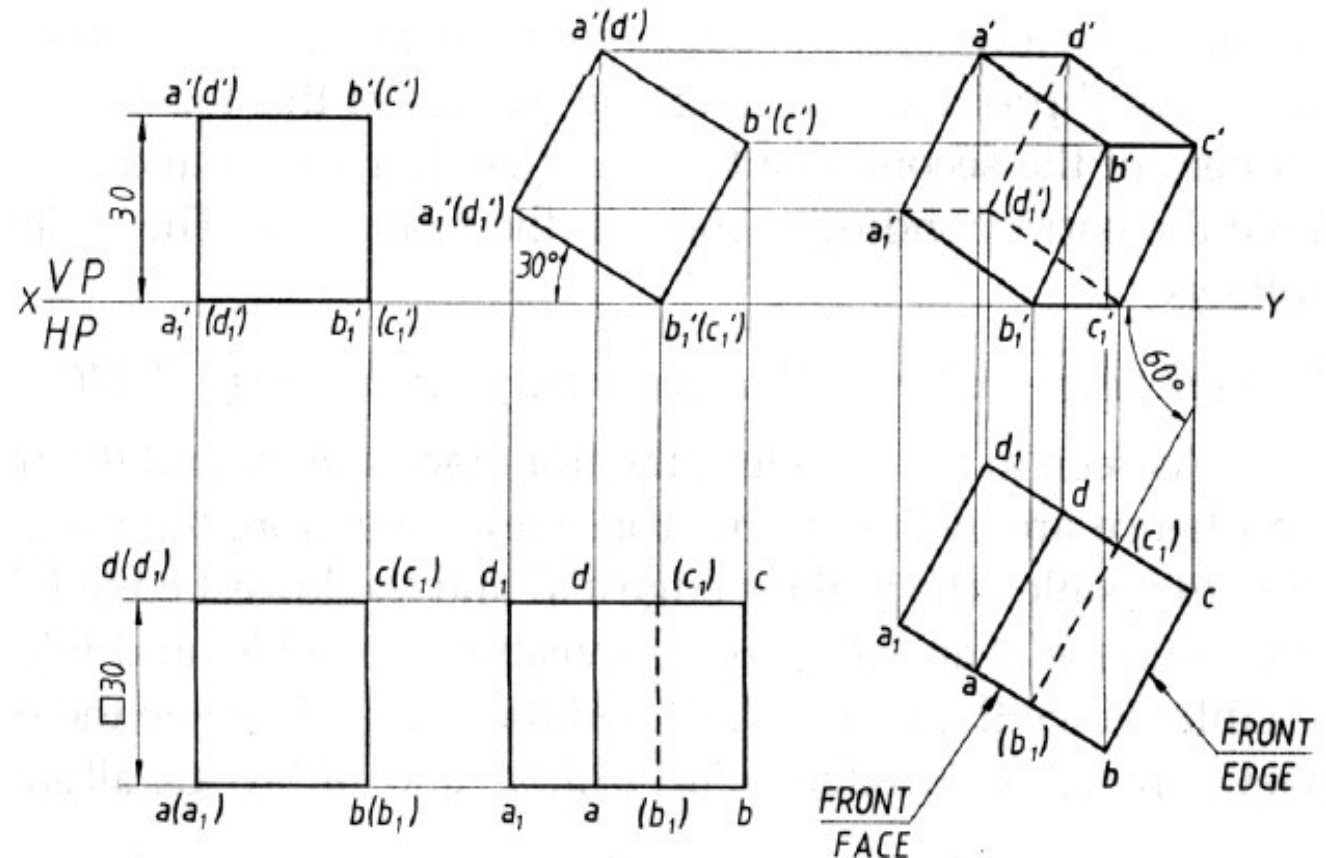
These projectors intersect each other to give the corresponding corners in the top view



Problem

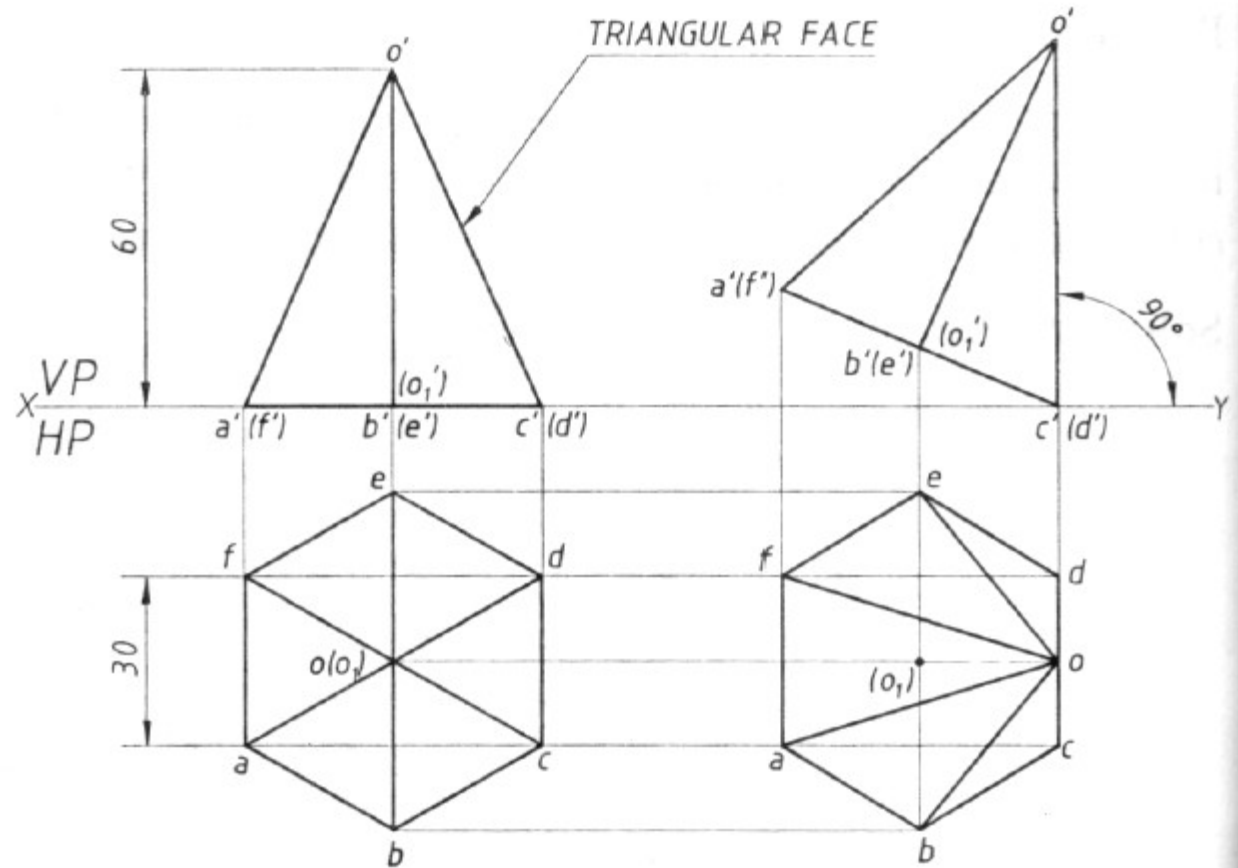
A cube of 30 mm sides rests with one of its edges on HP such that one of the square faces containing that edge is inclined at 30° to HP and the edge on which it rests being inclined to 60° to VP. Draw its projections.

Solution



A hexagonal pyramid has an altitude of 60 mm and side base 30mm. The pyramid rests on one of its side of the base on HP such that the triangular face containing that side is perpendicular to HP. Draw the front and top views.

Solution



Projections of solids with axes inclined to one of the reference planes and parallel to the other

When a solid has its axis inclined to one plane and parallel to the other, its projections are drawn in two stages.

a. In the initial stage, the solid is assumed to be in simple position, i.e. its axis perpendicular to one of the planes.

If the axis is to be inclined to the ground, i.e. the H.P., it is assumed to be perpendicular to the H.P. in the initial stage.

Similarly, if the axis is to be inclined to the V.P., it is kept perpendicular to the V.P. in the initial stage.

Moreover

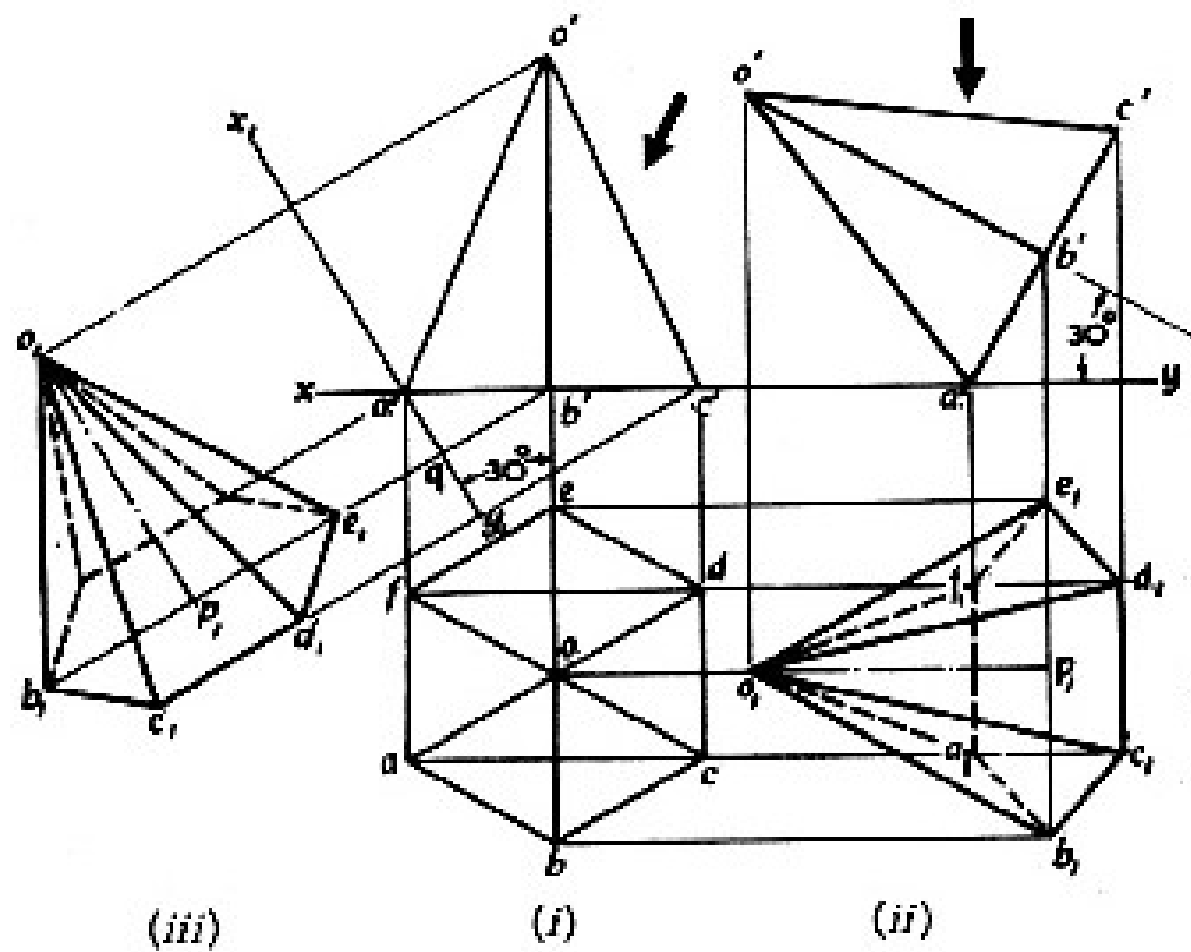
i. If the solid has an edge of its base parallel to the H.P. or in the H.P. or on the ground, that edge should be kept perpendicular to the V.P.; if the edge of the base is parallel to the V.P. or in the V.P., it should be kept perpendicular to the H.P.

ii. If the solid has a corner of its base in the H.P. or on the ground, the sides of the base containing that corner should be kept equally inclined to the V.P.; if the corner is in the V.P., they should be kept equally inclined to the H.P.

Projections of Solids with axis inclined to H.P and parallel to V.P

Problem 6.

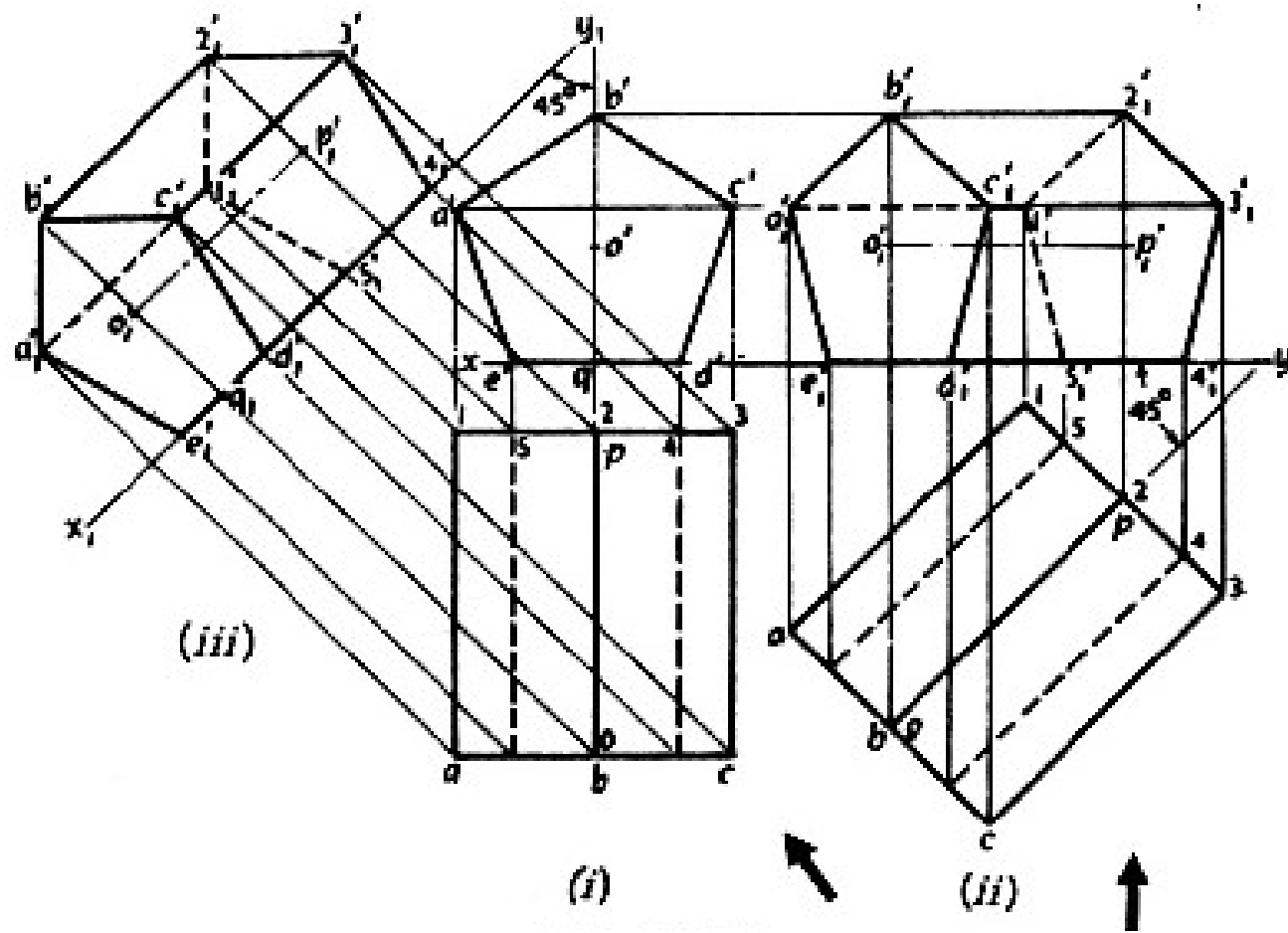
A hexagonal pyramid, base 25 mm side and axis 50 mm long, has an edge of its base on the ground. Its axis is inclined at 30° to the ground and parallel to the V.P. Draw its projections.



Projections of Solids with axis inclined to V.P and parallel to H.P

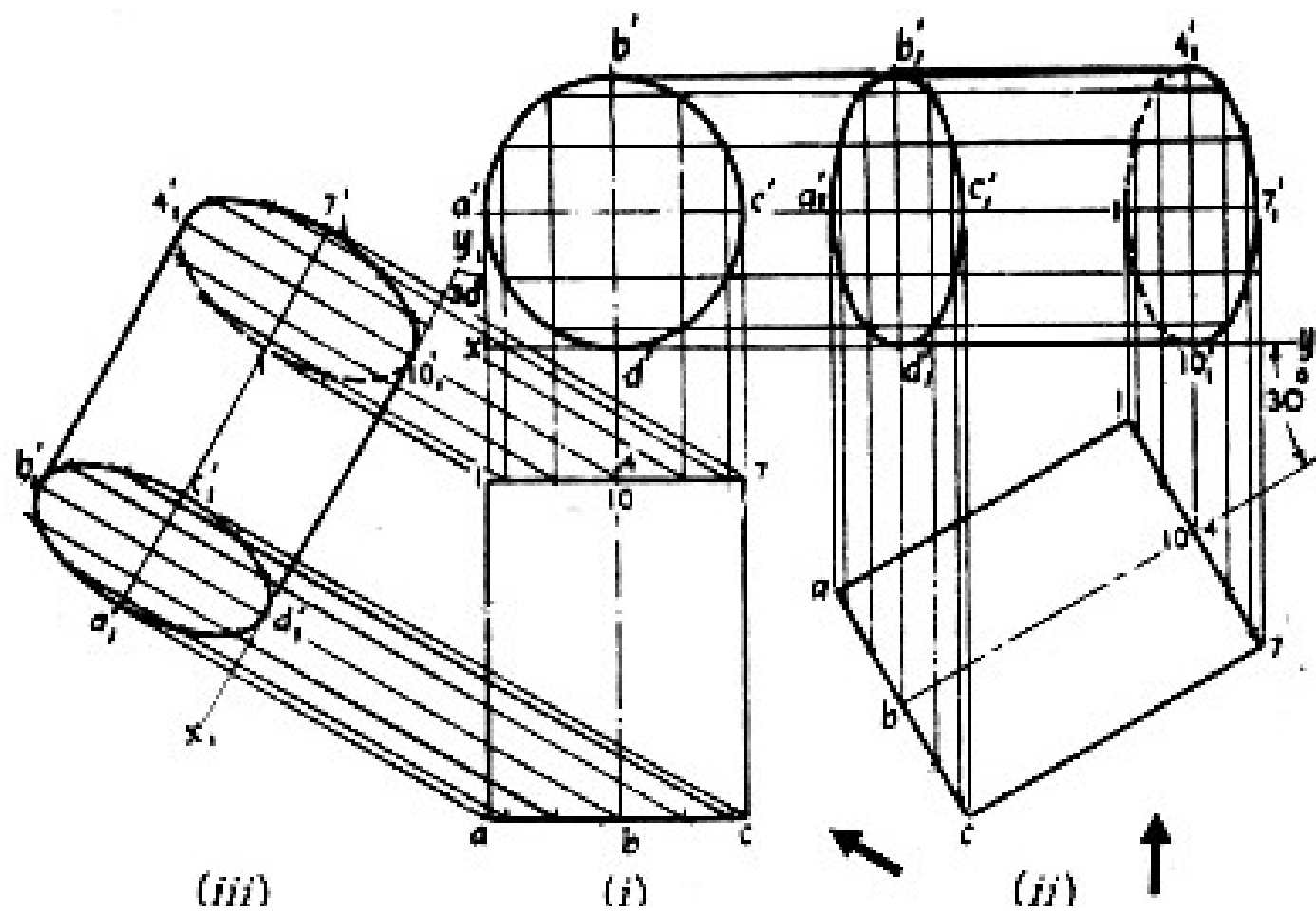
Problem 7 .

Draw the projections of a pentagonal prism, base 25 mm side and axis 50 mm long, resting on one of its rectangular faces on the H.P., with the axis inclined at 45° to the V.P.



Problem 8

Draw the projections of a cylinder 75 mm diameter and 100 mm long, lying on the ground with its axis inclined at 30° to the V.P. and parallel to the ground.



Projections of solids with axes inclined to both the H.P. and the V.P.

The projections of a solid with its axis inclined to both the planes are drawn in three stages:

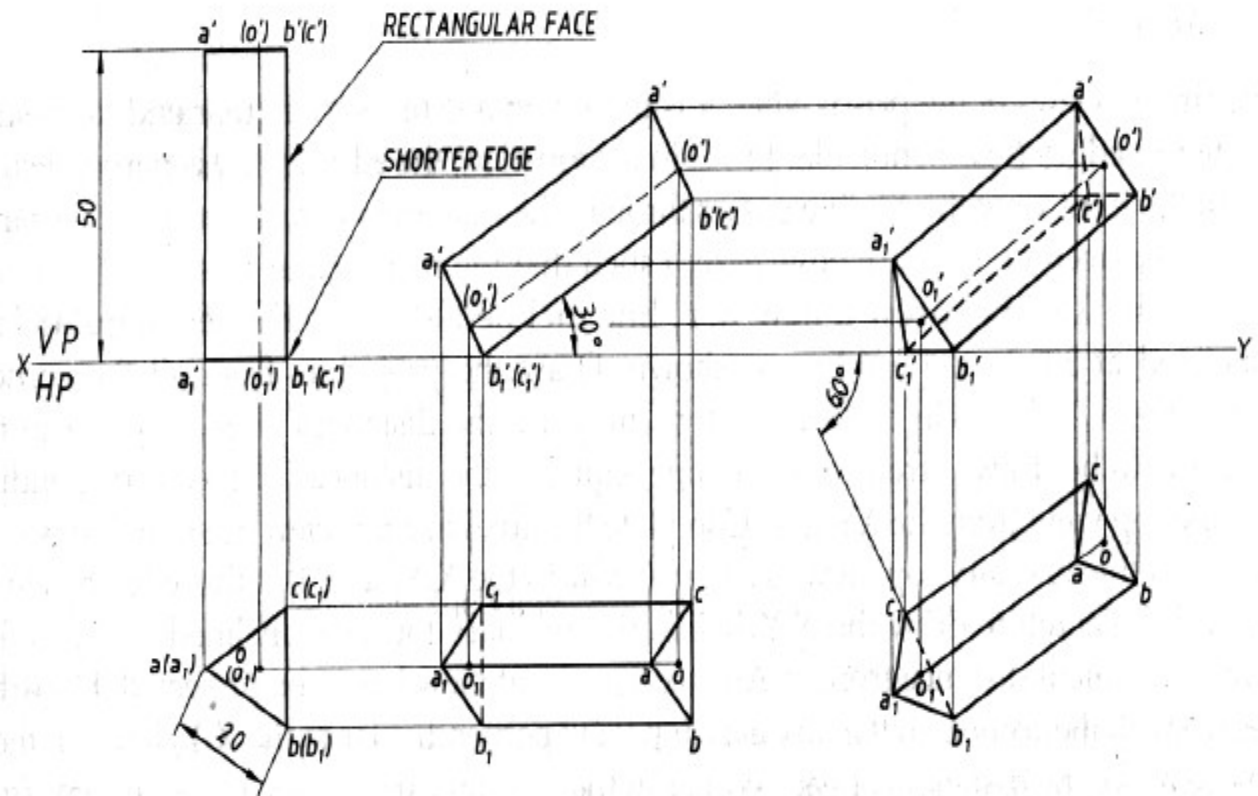
- i. Simple position
- ii. Axis inclined to one plane and parallel to the other
- iii. Final position

The second and final positions may be obtained either by the alteration of the positions of the solid, i.e., the views, or by the alteration of reference lines.

An equilateral triangular prism 20 mm side of base and 50 mm long rests with one of its shorter edges on HP such that the rectangular face containing the edge on which the prism rests is inclined at 30° to HP. The edge on which prism rests is inclined at 60° to VP. Draw its projections.

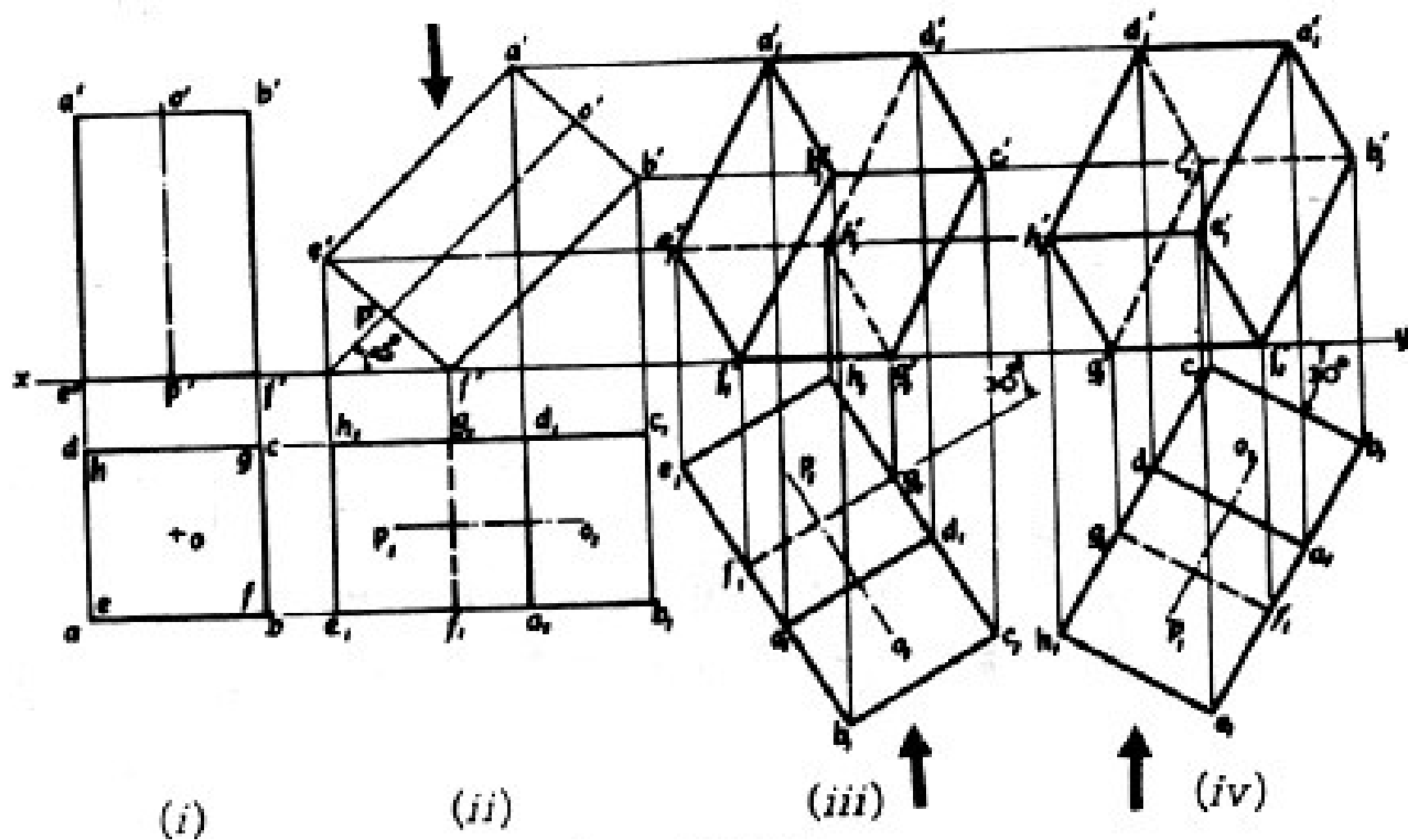
Solution:

The prism rests with one of its shorter edges, i.e., triangular or base edge on HP such that the rectangular face containing that edge is inclined at 30° to HP.



Problem 9

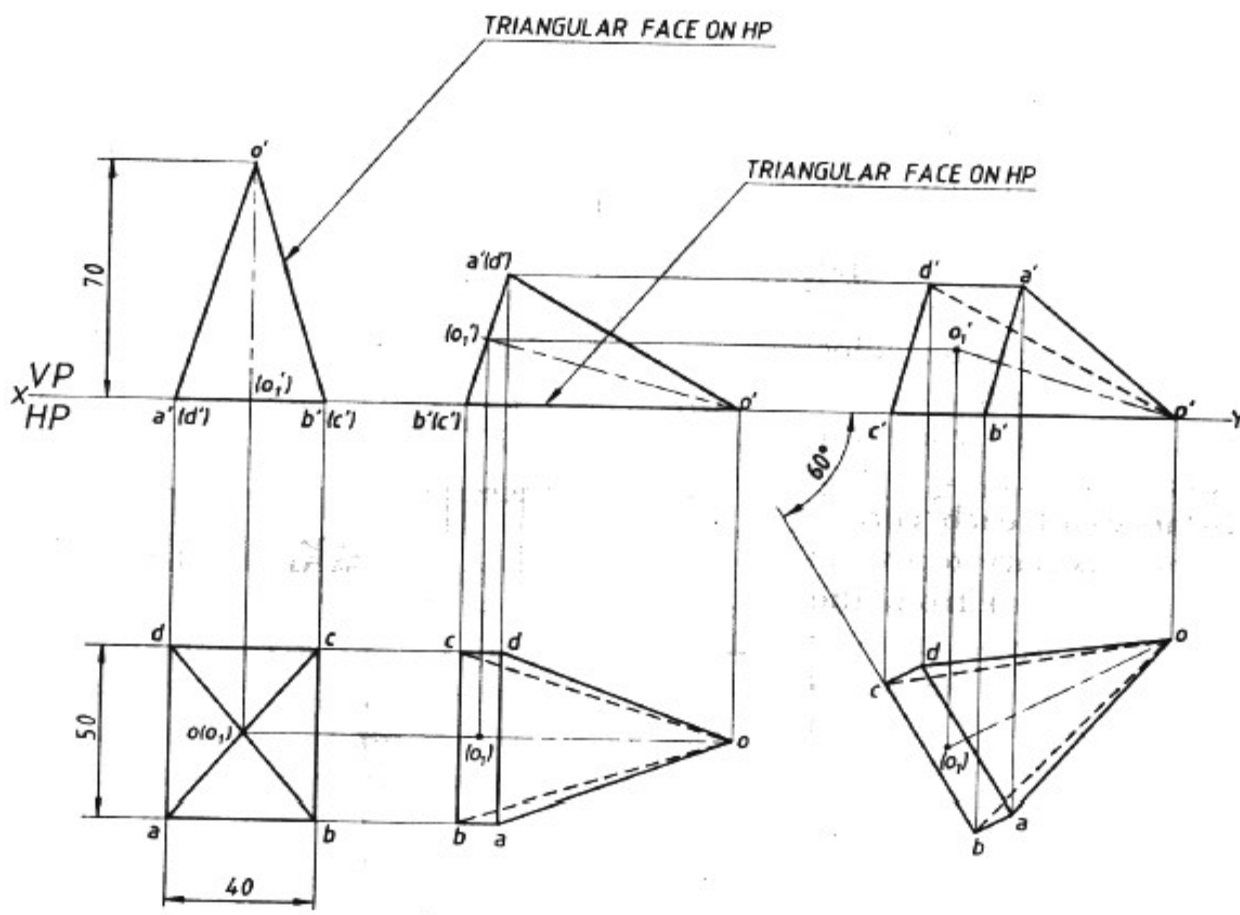
A square prism, base 40 mm side and height 65 mm, has its axis inclined at 45 degrees to the H.P. and has an edge of its base, on the H.P. and inclined at 30 degrees to the V.P. Draw its projections.



Problem

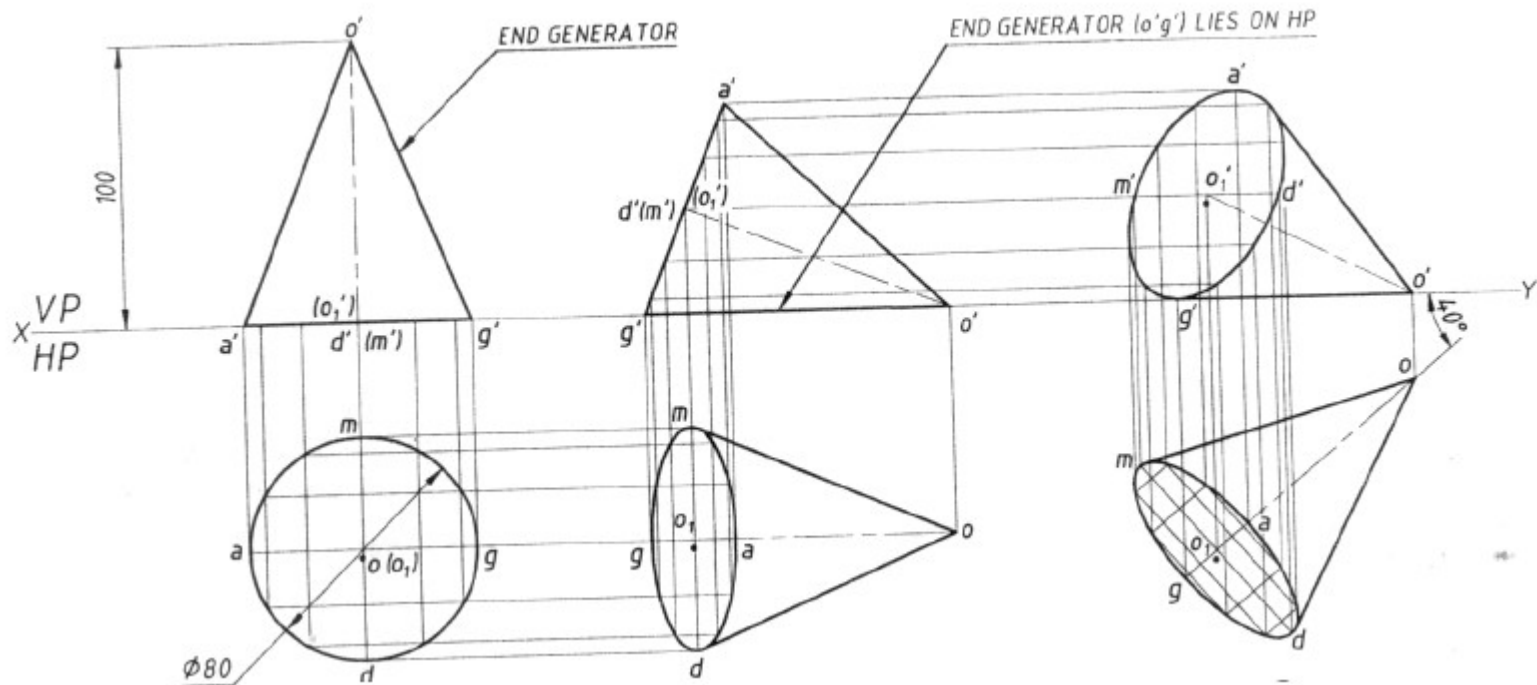
Draw the top and front views of a rectangular pyramid of sides of base 40x 50 mm and height 70 mm when it lies on one of its larger triangular faces on HP. The longer edge of the base of the triangular face lying on HP is inclined at 60° to VP in the top view with the apex of the pyramid being nearer to VP.

Solution



A cone of base 80 mm diameter and height 100 mm lies with one of its generators on HP and the axis appears to be inclined to VP at an angle of 40° in the top view. Draw its top and front views.

Solution



Draw the top and the front views of a right circular cylinder of base 45 mm diameter and 60 mm long when it lies on HP such that its axis is inclined at 35° to HP and the axis appears to be perpendicular to VP in the top view.

Solution

